

QUESTIONS 31 / 31 completed

Arithmetic

ARITHMETIC

Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.

PROGRAM EDITOR

Java

Run

Save

INPUT

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int a=in.nextInt();
6         int b=in.nextInt();
7         System.out.println(a+b);
8         System.out.println(a-b);
9         System.out.println(a*b);
10        System.out.println(a/b);
11    }
12 }
13 }
```

OUTPUT

30

QUESTIONS 31 / 31 completed

MAX

MAX

Write a Java program to find the maximum of three numbers using conditional statements.

PROGRAM EDITOR

```

1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] main){
4         Scanner ln=new Scanner(System.in);
5         int a=ln.nextInt();
6         int b=ln.nextInt();
7         int c=ln.nextInt();
8         if(a>b && a>c){
9             System.out.print(a);
10        }else if(b>a && b>c){
11            System.out.print(b);
12        }else{
13            System.out.print(c);
14        }
15    }
16 }
17 }
    
```

Java

Run

Save

OUTPUT

QUESTIONS 31 / 31 completed

Leap

LEAP

Write a Java program to check whether a given year is a leap year or not using conditional statements.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int n=in.nextInt();
6         if(n%400==0){
7             System.out.print("Leap year");
8         }else if(n%100 ==0){
9             System.out.print("Not Leap year");
10        }else if(n%4==0){
11            System.out.print("Leap year");
12        }else{
13            System.out.print("not a valid input");
14        }
15    }
16 }
```

Java

Run

Save

INPUT

2024

OUTPUT

Leap year

QUESTIONS 31 / 31 completed

Fibonacci

FIBONACCI

Write a java program to print the Fibonacci series up to a given number using loops.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int n=in.nextInt();
6         int first=0;
7         int second=1;
8         System.out.print(first + " ");
9         for(int i=1;i<=n;i++){
10             int next=first+second;
11             System.out.print(next + " ");
12             first=second;
13             second=next;
14         }
15
16
17
18
19 }
20 }
```

OUTPUT

0 1 2 3 5 8

Java

v

Run

Save

INPUT

5

QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int n=in.nextInt();
6         int sum=1;
7         for(int i=1;i<=n;i++){
8             sum=sum*i;
9         }
10        System.out.print(sum);
11    }
12 }
```

java

v

Run

Save

INPUT

OUTPUT

24

QUESTIONS 31 / 31 completed

RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops.

```

**
***
****
*****

```

PROGRAM EDITOR

```

1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int n=in.nextInt();
6         for(int i=0;i<n;i++){
7             for(int j=0;j<=i;j++){
8                 System.out.print("*");
9             }
10            System.out.println();
11        }
12    }
13 }
14 }

```

.java

Run

Save

INPUT

4

OUTPUT

QUESTIONS 31 / 31 completed

Search

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int arr[]=new int[5];
6         for(int i=0;i<5;i++){
7             arr[i]=in.nextInt();
8         }
9         int tar=in.nextInt();
10        int left=0;
11        int right=4;
12        while(left<right){
13            int mid=(right+left)/2;
14            if(arr[mid]==tar){
15                System.out.print(mid);
16            }
17            break;
18        }else if(arr[mid]>tar){
19            right=mid-1;
20        }else{
21            left=mid+1;
22        }
```

OUTPUT

2

MacBook Air

QUESTIONS 31 / 31 completed

Search

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

PROGRAM EDITOR

```

1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int arr[]=new int[5];
6         for(int i=0;i<5;i++){
7             arr[i]=in.nextInt();
8         }
9         int tar=in.nextInt();
10        int left=0;
11        int right=4;
12        while(left<right){
13            int mid=(right+left)/2;
14            if(arr[mid]==tar){
15                System.out.print(mid);
16                break;
17            }else if(arr[mid]>tar){
18                right=mid-1;
19            }else{
20                left=mid+1;
21        }
22    }
23 }

```

OUTPUT

2

MacBook Air

QUESTIONS 31 / 31 completed

Prime

PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int n=in.nextInt();
6         boolean isprime=true;
7         if(n<=1){
8             isprime=false;
9         }else{
10            for(int i=2;i<n;i++){
11                if(n%i==0){
12                    isprime=false;
13                    break;
14                }
15            }
16        }
17        if(isprime){
18            System.out.print("prime");
19        }else{
20            System.out.print("Not Prime");
21        }
22    }
23 }
```

OUTPUT

Not Prime

MacBook Air

QUESTIONS 31 / 31 completed

Sort

SORT

Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main{
3     public static void selectionsort(int[] arr){
4         int n=arr.length;
5         for(int i=0;i<n-1;i++){
6             int min=i;
7             for(int j=i+1;j<n;j++){
8                 if(arr[j]<arr[min]){
9                     min=j;
10                }
11            }
12            int temp=arr[min];
13            arr[min]=arr[i];
14            arr[i]=temp;
15        }
16    }
17    public static void main(String[] args){
18        Scanner sc=new Scanner(System.in);
19        int arr[]=new int[4];
20        for(int i=0;i<4;i++){
21            arr[i]=sc.nextInt();
```

OUTPUT

10 20 30 40

Java

Run

Save

INPUT

20 30 10 4

QUESTIONS 31 / 31 completed

Table

TABLE

Write a Java program to print the multiplication table of a given number using loops.

PROGRAM EDITOR

```
1 import java.util.Scanner;  
2 public class Main{  
3     public static void main(String[] args){  
4         Scanner n=new Scanner(System.in);  
5         int a=n.nextInt();  
6         for(int i=1;i<=10;i++){  
7             System.out.println(a*i);  
8         }  
9     }  
10 }
```

Java

Run

Save

INPUT

3

OUTPUT

3

MacBook Air

QUESTIONS 31 / 31 completed

Table

TABLE

Write a Java program to print the multiplication table of a given number using loops.

PROGRAM EDITOR

```
1 import java.util.Scanner;  
2 public class Main{  
3     public static void main(String[] args){  
4         Scanner n=new Scanner(System.in);  
5         int a=n.nextInt();  
6         for(int i=1;i<=10;i++){  
7             System.out.println(a*i);  
8         }  
9     }  
10 }
```

Java

Run

Save

INPUT

3

OUTPUT

3

MacBook Air

QUESTIONS 31 / 31 completed

LT-Pattern

LT-PATTERN

Write a Java program to print the following pattern using loops. *****

```
*****
**
*
```

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int n=in.nextInt();
6         for(int i=0;i<n;i++){
7             for(int j=0;j<n-i;j++){
8                 System.out.print("*");
9             }
10            System.out.println();
11        }
12    }
13 }
```

OUTPUT

Java

v

Run

Save

INPUT

5

QUESTIONS 31 / 31 completed

LT-Pattern

LT-PATTERN

Write a Java program to print the following pattern using loops. *****

**
*
.

PROGRAM EDITOR

```
1 import java.util.Scanner;  
2 public class Main{  
3     public static void main(String[] args){  
4         Scanner in=new Scanner(System.in);  
5         int n=in.nextInt();  
6         for(int i=0;i<n;i++){  
7             for(int j=0;j<n-i;j++){  
8                 System.out.print("*");  
9             }  
10            System.out.println();  
11        }  
12    }  
13 }
```

OUTPUT

Java

v

Run

Save

INPUT

5

INPUT

D T O B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int n=in.nextInt();
6         String binary=Integer.toString(n);
7         System.out.print(binary);
8     }
9 }
```

OUTPUT

10

QUESTIONS 31 / 31 completed

Sum

SUM

Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int n=in.nextInt();
6         int sum=0;
7         while(n!=0){
8             int rem=n%10;
9             sum+=rem;
10            n=n/10;
11        }
12        System.out.print(sum);
13    }
14 }
```

Java

v

Run

Save

INPUT

123

OUTPUT

6

QUESTIONS 31 / 31 completed

Product

v

PRODUCT

Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int n=in.nextInt();
6         int sum=1;
7         while(n!=0){
8             int rem=n%10;
9             sum*=rem;
10            n=n/10;
11        }
12        System.out.print(sum);
13    }
14 }
```

Java

v

Run

Save

INPUT

223

OUTPUT

12

QUESTIONS 31 / 31 completed

Product

PRODUCT

Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int n=in.nextInt();
6         int sum=1;
7         while(n!=0){
8             int rem=n%10;
9             sum*=rem;
10            n=n/10;
11        }
12        System.out.print(sum);
13    }
14 }
```

Java

Run

Save

INPUT

223

OUTPUT

12

QUESTIONS 31 / 31 completed

Area Circle

AREA CIRCLE

Write a Java program to calculate the area of a circle using the radius input by the user.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int n=in.nextInt();
6         double pie=3.14;
7         System.out.print(pie*n*n);
8     }
9 }
```

Java

Run

Save

INPUT

7

OUTPUT

153.86

QUESTIONS 31/31 completed

Shpere

SHPERE

Write a Java program to calculate the volume of a sphere using the radius input by the user.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;  
2 public class Main{  
3     public static void main(String[] args){  
4         Scanner in= new Scanner(System.in);  
5         int n=in.nextInt();  
6         double pi=3.14;  
7         double result=(4/3)*pi*n*n*n;  
8         System.out.print(result);  
9     }  
10 }
```

OUTPUT

84.78

3

INPUT

QUESTIONS 31 / 31 completed

Palindrome ✓

PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     static boolean ispalindrome(String str,int left,int right){
4         return (left<=right)|| (str.charAt(left)==str.charAt(right) && ispalindrome(s
5         )
6     }
7     public static void main(String[] args){
8         Scanner in=new Scanner(System.in);
9         String str=in.nextLine();
10        System.out.println(ispalindrome(str,0,str.length()-1));
11    }
12 }
```

OUTPUT

false

INPUT

sumanth

QUESTIONS 31 / 31 completed

Armstrong

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main{
3     static int armstrong(int n,int digits){
4         return n==070:(int)Math.pow(n%10,digits)+armstrong(n/10,digits);
5     }
6     public static void main(String[] args){
7         Scanner in=new Scanner(System.in);
8         int n=in.nextInt();
9         int digits=String.valueOf(n).length();
10        System.out.print(armstrong(n,digits)==n?"yes":"no");
11    }
12 }
```

Java

Run

Save

INPUT

123

OUTPUT

no

QUESTIONS 31 / 31 completed

Armstrong

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main{
3     static int armstrong(int n,int digits){
4         return n==070:(int)Math.pow(n%10,digits)+armstrong(n/10,digits);
5     }
6     public static void main(String[] args){
7         Scanner in=new Scanner(System.in);
8         int n=in.nextInt();
9         int digits=String.valueOf(n).length();
10        System.out.print(armstrong(n,digits)==n?"yes":"no");
11    }
12 }
```

Java

Run

Save

INPUT

123

OUTPUT

no

QUESTIONS 31/31 completed

Perfect

PERFECT

Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main{
3     static int perfect(int n,int i){
4         return i>n/2?0:(n%i==0)?perfect(n,i+1):0;
5     }
6     public static void main(String[] args){
7         Scanner in=new Scanner(System.in);
8         int n=in.nextInt();
9         System.out.print(perfect(n,1)+"yes":"no");
10    }
11 }
```

Java

Run

Save

INPUT

6

OUTPUT

yes

QUESTIONS 31 / 31 completed

Max Array

MAX ARRAY

Write a Java program to find the largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int n=in.nextInt();
6         int arr[]=new int[n];
7         for(int i=0;i<n;i++){
8             arr[i]=in.nextInt();
9         }
10        int max=arr[0];
11        for(int i=0;i<n;i++){
12            if(arr[i]>max){
13                max=arr[i];
14            }
15        }
16        System.out.print(max);
17    }
18 }
19 }
```

OUTPUT

43

QUESTIONS 31 / 31 completed

2nd Max

2ND MAX

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int n=in.nextInt();
6         int arr[]=new int[n];
7         for(int i=0;i<n;i++){
8             arr[i]=in.nextInt();
9         }
10        int max=arr[0];
11        int maxs=arr[0];
12        for(int i=0;i<n;i++){
13            if(arr[i]>max){
14                maxs=max;
15                max=arr[i];
16            }else if(arr[i]>maxs && arr[i]!=max){
17                maxs=arr[i];
18            }
19        }
20        System.out.print(maxs);
21    }
```

OUTPUT

22

Java

Run

Save

INPUT

4 21 43 22 12

QUESTIONS 31 / 31 completed

2nd Max

2ND MAX

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int n=in.nextInt();
6         int arr[]=new int[n];
7         for(int i=0;i<n;i++){
8             arr[i]=in.nextInt();
9         }
10        int max=arr[0];
11        int maxs=arr[0];
12        for(int i=0;i<n;i++){
13            if(arr[i]>max){
14                maxs=max;
15                max=arr[i];
16            }else if(arr[i]>maxs && arr[i]!=max){
17                maxs=arr[i];
18            }
19        }
20        System.out.print(maxs);
21    }
```

OUTPUT

22

Java

v

Run

Save

INPUT

4 21 43 22 12

QUESTIONS 31 / 31 completed

2nd Min

2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

INPUT

4 23 21 22 24

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int n=in.nextInt();
6         int arr[]=new int[n];
7         for(int i=0;i<n;i++){
8             arr[i]=in.nextInt();
9         }
10        int min=arr[0];
11        int mins=arr[0];
12        for(int i=0;i<n;i++){
13            if(arr[i]<min){mins=min;
14                min=arr[i];
15            } else if(arr[i]<mins && arr[i]!=min){
16                mins=arr[i];
17            }
18        }
19        System.out.print(mins);
20    }
21 }
```

OUTPUT

22

QUESTIONS 31 / 31 completed

2nd Min

2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

INPUT

4 23 21 22 24

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int n=in.nextInt();
6         int arr[]=new int[n];
7         for(int i=0;i<n;i++){
8             arr[i]=in.nextInt();
9         }
10        int min=arr[0];
11        int mins=arr[0];
12        for(int i=0;i<n;i++){
13            if(arr[i]<min){mins=min;
14                min=arr[i];
15            } else if(arr[i]<mins && arr[i]!=min){
16                mins=arr[i];
17            }
18        }
19        System.out.print(mins);
20    }
21 }
```

OUTPUT

22

QUESTIONS 31 / 31 completed

Sort-1

SORT-1

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner in=new Scanner(System.in);
5         int n=in.nextInt();
6         int arr[]=new int[n];
7         for(int i=0;i<n;i++){
8             arr[i]=in.nextInt();
9         }
10        for(int i=1;i<n;i++){
11            int key=arr[i];
12            int j=i-1;
13            while(j>=0 && arr[j]>key){
14                arr[j+1]=arr[j];
15                j--;
16            }
17            arr[j+1]=key;
18        }
19        for(int i=0;i<n;i++){
20            System.out.print(arr[i]+" ");
21        }
```

OUTPUT

23 32 34 43

Java

v

Run

Save

INPUT

4 32 23 34 43